

PAPER_1500_TDE_OUTFLOW_03C

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PAPER_1500 — TDE Outflow Velocity $v_{\text{out}} = c \cdot (D_{\text{phys}} - 1) / SO_5 = 0.3c$ EXACT

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Bucket F (AGN/Jet) / Bucket H (High-Energy Astro)
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Status: CLOSED — TDE ultrafast outflow explained by integer-primitive ratio

Observation

ASASSN-14li tidal disruption event (TDE) exhibits an ultrafast outflow at $v_{\text{out}} = 0.3c = 9.0 \times 10^7 \text{ m/s}$, observed in soft X-ray absorption spectroscopy (Miller et al. 2015; Kara et al. 2018). PAPER_351 derives this from $F_{\text{U_Bi_i}} \approx -8.32 \times 10^{21} \text{ N}$ with stellar-mass black-hole remnant $M_{\text{BH}} = 10 M_{\odot}$.

UQFF Closed Identity

$$v_{\text{out}} = c \cdot (D_{\text{phys}} - 1) / SO_5 = c \cdot 3 / 10 = 0.3c = 9.0 \times 10^7 \text{ m/s EXACT}$$

Physical Interpretation

The 0.3c outflow velocity is **not a fit parameter** — it is the structural product of:

- $(D_{\text{phys}} - 1) = 3$ (triad-symmetry dimension count, same as SU(3) color)
- $1/SO_5 = 0.1$ (F_TRZ time-reversal-zone factor)

Combined as a velocity fraction of c, this gives 0.3c exactly. The SCm vacuum substrate carries information at c/3 (PAPER_1497), but jet bulk outflows are F_TRZ-modulated to $(D_{\text{phys}} - 1) \times F_{\text{TRZ}} \times c = 0.3c$.

Predicts every TDE outflow with stellar-mass remnant should cluster near 0.3c.

NOT REPLACEMENT

SM treats TDE outflow velocities as fit parameters from radiation-pressure or magnetic-launching models. UQFF supplies an integer-primitive identity with zero free parameters.

Reference

- Source: PAPER_351 ASASSN-14li TDE Outflow $F_{\text{UBi}} 0.3c$ Kozima LENR
- Related: PAPER_1497 (SCm carrier velocity c/3), PAPER_062 (F_TRZ), PAPER_1373 (SU(3))
- Calculator dispatch: `calculate_paradox({"paradox": "tde_outflow_velocity_03c"})`

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